





goals





goals



winners





goals



winners



stopped
penalties





goals



winners



stopped
penalties



Cases



goals



winners



stopped
penalties



Cases

Variables

Cases **Vs.** Variables

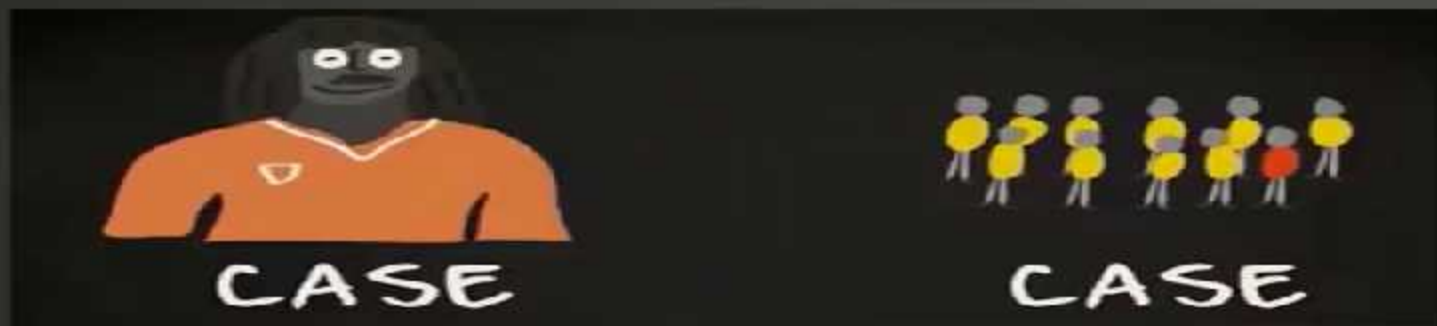
Cases Vs. Variables



Cases Vs. Variables



Cases Vs. Variables



Cases Vs. Variables



Cases Vs. Variables



TEAM 1



TEAM 2



TEAM 3

"..."

VARIABLE →

city where the team is based

Barcelona

Madrid

Valencia

Sevilla

CONSTANT →

country where the team is based

Spain

Spain

Spain

Spain

Cases Vs. Variables

[illegible]

Level of Measurements

Level of Measurements

	nominal	Difference			
		+			

Level of Measurements

	Difference	Order		
nominal	+	-		

Level of Measurements

	Difference	Order		
nominal	+	-		
ordinal	+			

Level of Measurements

	Difference	Order		
nominal	+	-		
ordinal	+	+		

Level of Measurements

	Difference	Order	Similar intervals
nominal	+	-	-
ordinal	+	+	-

Level of Measurements

Categorical		Difference	Order	Similar intervals
	nominal	+	-	-
	ordinal	+	+	-

Level of Measurements

Categorical		Difference	Order	Similar intervals
	nominal	+	-	-
	ordinal	+	+	-
	interval	+	+	+

Level of Measurements

Categorical		Difference	Order	Similar intervals	Meaningful zero point
	nominal	+	-	-	-
	ordinal	+	+	-	-
	interval	+	+	+	-
	ratio	+	+	+	+

Level of Measurements

		Difference	Order	Similar intervals	Meaningful zero point
Categorical	nominal	+	-	-	-
	ordinal	+	+	-	-
Quantitative	interval	+	+	+	-
	ratio	+	+	+	+

Level of Measurements

		Difference	Order	Similar intervals	Meaningful zero point
Categorical	nominal	+	-	-	-
	ordinal	+	+	-	-
Quantitative	interval	+	+	+	-
	ratio	+	+	+	+

Continuous

Discrete

Level of Measurements

		Difference	Order	Similar intervals	Meaningful zero point
Categorical	nominal	+	-	-	-
	ordinal	+	+	-	-
Quantitative	interval	+	+	+	-
	ratio	+	+	+	+

Continuous

170

Discrete

Level of Measurements

		Difference	Order	Similar intervals	Meaningful zero point
Categorical	nominal	+	-	-	-
	ordinal	+	+	-	-
Quantitative	interval	+	+	+	-
	ratio	+	+	+	+

Continuous

170

171

Discrete

Level of Measurements

		Difference	Order	Similar intervals	Meaningful zero point
Categorical	nominal	+	-	-	-
	ordinal	+	+	-	-
Quantitative	interval	+	+	+	-
	ratio	+	+	+	+

Continuous

170

171

170.2461

Discrete

Level of Measurements

		Difference	Order	Similar intervals	Meaningful zero point
Categorical	nominal	+	-	-	-
	ordinal	+	+	-	-
Quantitative	interval	+	+	+	-
	ratio	+	+	+	+

Continuous

170

171

170.2461

Discrete

5

4



Primera División



Primera División

CASES





Primera División

CASES



VARIABLES

Age

Weight

Goals scored

Team membership

Hair color

VARIABLES

Matrix	Age	Weight	Goals scored	Team membership	Hair color
Player 1	18	72.6	0	Real Zaragoza	Blond
Player 2	21	71.4	0	Real Betis	Black
Player 3	26	74.8	8	Sevilla	Black
Player 4	22	76.8	12	Barcelona	Black
Player 5	22	74.1	17	Valencia	Other
Player 6	27	78.9	3	Real Sociedad	Other
Player 7	30	80.3	2	Real Madrid	Blond
Player 8	24	73.3	1	Athletic Bilbao	Brown
Player 9	23	76.9	5	Valencia	Brown
Player 400	26	77.2	0	Athletic Madrid	Other

MATRIX	VARIABLES				
	Age	Weight	Goals scored	Team membership	Hair color
Player 1	28	75 kg	10	Local Community	Brown
Player 2	32	78 kg	15	Local Club A	Brown
Player 3	25	74 kg	8	Local Club B	Brown
Player 4	30	76 kg	12	Local Club C	Brown
Player 5	31	79 kg	17	Local Club D	Brown
Player 6	27	73 kg	5	Local Club E	Brown
Player 7	33	80 kg	9	Local Club F	Brown
Player 8	29	75 kg	7	Local Club G	Brown
Player 9	26	72 kg	6	Local Club H	Brown
Player 10	34	81 kg	11	Local Club I	Brown

MATRIX

VARIABLES

	Age	Weight	Goals scored	Team membership	Hair color
Player 1	19	72.6	0	Real Zaragoza	Blond
Player 2	21	71.9	0	Real Betis	Black
Player 3	26	79.8	8	Sevilla	Black
Player 4	22	76.8	12	Barcelona	Black
Player 5	22	79.1	17	Valencia	Other
Player 6	27	78.9	3	Real Sociedad	Other
Player 7	30	80.3	2	Real Madrid	Blond
Player 8	24	75.3	1	Atletico Bilbao	Brown
Player 9	23	76.9	5	Valencia	Brown
Player 400	26	77.2	0	Atletico Madrid	Other

MATRIX

Player 1
Player 2
Player 3
Player 4
Player 5
Player 6
Player 7
Player 8
Player 9

Player 400

Age

19
21
26
22
22
27
30
24
23

25

Weight

72.6
71.4
79.8
76.8
74.1
78.9
80.3
75.3
76.9

77.2

Goals scored

0
0
8
12
17
3
2
1
5

0

VARIABLES

Team membership

Real Zaragoza
Real Betis
Sevilla
Barcelona
Valencia
Real Sociedad
Real Madrid
Athletic Bilbao
Valencia

Athletic Madrid

Hair color

Blond
Black
Black
Black
Other
Other
Blond
Brown
Brown

Other

MATRIX

Player 1
Player 2
Player 3
Player 4
Player 5
Player 6
Player 7
Player 8
Player 9

Player 400

Age

18
21
26
22
22
27
30
24
23

25

Weight

72.6
71.4
79.8
76.8
74.1
78.9
80.3
75.3
76.9

77.2

Goals scored

0
0
8
12
17
3
2
1
5

0

VARIABLES

Team membership

Real Zaragoza
Real Betis
Sevilla
Barcelona
Valencia
Real Sociedad
Real Madrid
Athletic Bilbao
Valencia

Athletic Madrid

Hair color

Blond
Black
Black
Black
Other
Other
Blond
Brown
Brown

Other



VARIABLES

MATRIX	Age	Weight	Goals scored	Team membership	Hair color
Player 1	18	72.6	0	Real Zaragoza	Blond
Player 2	21	71.4	0	Real Betis	Black
Player 3	26	79.8	8	Sevilla	Black
Player 4	22	76.8	12	Barcelona	Black
Player 5	22	74.1	17	Valencia	Other
Player 6	27	78.9	3	Real Sociedad	Other
Player 7	30	80.3	2	Real Madrid	Blond
Player 8	29	73.3	1	Athletic Bilbao	Brown
Player 9	23	76.9	5	Valencia	Brown
Player 400	26	77.2	0	Athletic Madrid	Other

MATRIX

Player 1
Player 2
Player 3
Player 4
Player 5
Player 6
Player 7
Player 8
Player 9

Player 400

Age

19

21

26

22

22

27

30

24

23

26

Weight

72.6

71.4

79.8

76.8

79.1

78.9

80.3

73.3

76.9

77.2

Goals scored

0

0

8

12

17

3

2

1

5

0

VARIABLES

Team membership

Real Zaragoza

Real Betis

Sevilla

Barcelona

Valencia

Real Sociedad

Real Madrid

Athletic Bilbao

Valencia

Athletic Madrid

Hair color

Blond

Black

Black

Black

Other

Other

Blond

Brown

Brown

Other

MATRIX

VARIABLES



	Age	Weight	Goals scored	Team membership	Hair color
Player 1	19	72.5	0	Real Zaragoza	Blond
Player 2	21	71.4	0	Real Betis	Black
Player 3	26	79.8	8	Sevilla	Black
Player 4	22	76.8	12	Barcelona	Black
Player 5	23	74.1	17	Valencia	Other
Player 6	27	78.9	3	Real Sociedad	Other
Player 7	30	80.3	2	Real Madrid	Blond
Player 8	24	75.3	1	Athletic Bilbao	Brown
Player 9	23	76.9	5	Valencia	Brown
Player 400	26	77.2	0	Athletic Madrid	Other

MATRIX

OBSERVATIONS

VARIABLES

	Age	Weight	Goals scored	Team membership	Hair color
Player 1	18	72.6	0	Real Zaragoza	Blond
Player 2	21	71.4	0	Real Betis	Black
Player 3	26	79.8	8	Sevilla	Black
Player 4	22	76.8	12	Barcelona	Black
Player 5	22	79.1	17	Valencia	Other
Player 6	27	77.9	3	Real Sociedad	Other
Player 7	30	80.3	2	Real Madrid	Blond
Player 8	24	75.3	1	Athletic Bilbao	Brown
Player 9	23	76.9	5	Valencia	Brown
Player 400	26	77.2	0	Athletic Madrid	Other

MATRIX	OBSERVATIONS		VARIABLES		
	Age	Weight	Goals scored	Team membership	Hair color
Player 1	18	72.6	0	Real Zaragoza	Blond
Player 2	21	71.4	0	Real Betis	Black
Player 3	26	79.8	8	Sevilla	Black
Player 4	22	76.8	12	Barcelona	Black
Player 5	22	79.1	17	Valencia	Other
Player 6	27	78.9	3	Real Sociedad	Other
Player 7	30	80.3	2	Real Madrid	Blond
Player 8	24	75.3	1	Athletic Bilbao	Brown
Player 9	23	76.9	5	Valencia	Brown
Player 400	26	77.2	0	Athletic Madrid	Other

DATA MATRIX

VARIABLES

CASES

	Age	Weight	Goals scored	Team membership	Hair color
Player 1	18	72.6	0	Real Zaragoza	Blond
Player 2	21	71.4	0	Real Betis	Black
Player 3	26	74.8	8	Sevilla	Black
Player 4	22	76.8	12	Barcelona	Black
Player 5	22	74.1	17	Valencia	Other
Player 6	27	78.9	3	Real Sociedad	Other
Player 7	30	80.3	2	Real Madrid	Blond
Player 8	24	73.3	1	Athletic Bilbao	Brown
Player 9	23	76.9	5	Valencia	Brown
...					
Player 400	26	77.2	0	Athletic Madrid	Other

CASES

DATA MATRIX

Player 1
Player 2
Player 3
Player 4
Player 5
Player 6
Player 7
Player 8
Player 9

Age

18
21
26
22
22
27
30
24
23

Weight

72.6
71.4
74.8
76.8
74.1
78.9
80.3
73.3
76.9

Goals scored

0
0
8
12
17
3
2
1
5

VARIABLES

Team membership

Real Zaragoza
Real Betis
Sevilla
Barcelona
Valencia
Real Sociedad
Real Madrid
Athletic Bilbao
Valencia

Hair color

Blond
Black
Black
Black
Other
Other
Blond
Brown
Brown

CASES

DATA MATRIX

VARIABLES

Age

Weight

Goals scored

Team membership

Hair color

Player 10

23

71.4

0

Real Betis

Black

Player 11

18

71.5

0

Real Zaragoza

Blond

Player 12

21

56.4

8

Sevilla

Black

Player 13

34

76.8

12

Barcelona

Black

Player 14

33

74.1

17

Valencia

Other

Player 15

44

78.9

3

Real Sociedad

Other

Player 16

27

80.3

2

Real Madrid

Blond

Player 17

24

72.6

1

Athletic Bilbao

Brown

Player 18

24

76.9

5

Valencia

Brown

Player 19

18

71.5

0

Real Zaragoza

Blond

Player 20

33

74.1

17

Valencia

Other

Player 21

44

78.9

3

Real Sociedad

Other

DATA MATRIX

VARIABLES

CASES

Age

Weight

Goals scored

Team membership

Hair color

Player 22

23

75.1

3

Valencia

Other

Player 23

22

76.3

5

Barcelona

Black

Player 24

26

-

0

Málaga

Black

Player 25

-

80.1

0

Villareal

Blond

Player 26

27

77.1

7

Getafe

Brown

DATA MATRIX

VARIABLES

CASES

Age

Weight

Goals scored

Team membership

Hair color

Player 22

23

75.1

3

Valencia

Other

Player 23

22

76.3

5

Barcelona

Black

Player 24

26

0

Málaga

Black

Player 25

80.1

0

Villareal

Blond

Player 26

27

77.1

7

Getafe

Brown

DATA MATRIX

VARIABLES

CASES

	Age	Weight	Goals scored	Team membership	Hair color
Player 22	23	75.1	3	Valencia	Other
Player 23	22	76.3	5	Barcelona	Black
Player 24	26		0	Málaga	Black
Player 25		80.1	0	Villareal	Blond
Player 26	27	77.1	7	Getafe	Brown

you might have to remove them

HVGE

Hair color			
Blond			
Brown			
Black			
other			
Total			

Hair color	Frequency		
Blond	76		
Brown	134		
Black	160		
other	30		
Total	400		

Hair color	Frequency	Percentage
Blond	76	19
Brown	134	33.5
Black	160	40
other	30	7.5
Total	400	100

Hair color

Frequency

Percentage

Blond

76

19

Brown

134

33.5

Black

160

40

other

30

7.5

Total

400

100

$$76/400 \times 100$$

Hair color	Frequency	Percentage	Cumulative percentage
Blond	76	19	19
Brown	134	33.5	52.5
Black	160	40	92.5
other	30	7.5	100
Total	400	100	

Hair color	Frequency	Percentage	Cumulative percentage
Blond	76	19	19
Brown	134	33.5	$19 + 33.5 \rightarrow 52.5$
Black	160	40	92.5
other	30	7.5	100
Total	400	100	

CATEGORICAL



Hair color	Frequency	Percentage	Cumulative percentage
Blond	76	19	19
Brown	134	33.5	52.5
Black	160	40	92.5
other	30	7.5	100
Total	400	100	

QUANTITATIVE



Weight (in kgs)	Frequency	Percentage
65.3	2	0.5
65.4	1	0.25
65.5	3	0.75
65.6	1	0.25
65.7	0	0
65.8	0	0
65.9	1	0.25
...		

QUANTITATIVE



Weight (in kgs)	Frequency	Percentage
65.3	2	0.5
65.4	1	0.25
65.5	3	0.75
65.6	1	0.25
65.7	0	0
65.8	0	0
65.9	1	0.25
...		

Interval

Weight (in kgs)	Frequency	Percentage
less than 60	8	2
60-69.9	69	17.25
70-79.9	273	68.25
80-89.9	45	11.25
90 and more	5	1.25
Total	400	100